TPCT'S TERNA ENGINEERING COLLEGE
Knowledge to Empower
An Autonomous Institute Affiliated to University of Mumbai | Approved by AICTE
DTE Maharashtra State | NAAC Accredited | NBA Accredited Programs

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2063301	Logical Reasoning	02	--	--	30	--	--	30	60	2

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2063301	Logical Reasoning	25	25	50	50	2	---	---	100

*** Course Engagement Component: Five marks integrated within IAT-I and IAT-II respectively**

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

Logical reasoning is a critical component of campus recruitment examinations conducted by leading IT, consulting, analytics, banking, and core engineering organizations. The course develops analytical thinking, problem-solving ability, pattern recognition, and decision-making skills. It provides students with a strong foundation required for aptitude tests, coding assessments, group discussions, and technical interviews.

Course Objectives:

Sr. No.	Course Objectives
1	Develop analytical and critical thinking abilities.
2	Improve problem-solving speed and accuracy.
3	Understand various logical reasoning techniques used in placement tests.
4	Analyze patterns, relationships, and sequences.
5	Strengthen decision-making and inference skills.
6	Build confidence for competitive examinations and campus recruitment processes.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Apply logical reasoning techniques to solve analytical problems.	L3 – Apply
2	Interpret relationships, patterns, and arrangements accurately.	L2 – Understand
3	Analyze statements, assumptions, and conclusions logically.	L4 – Analyze
4	Solve puzzles and data-based reasoning questions efficiently.	L3 – Apply
5	Evaluate decision-making situations using logical principles.	L5 – Evaluate
6	Demonstrate readiness for placement-oriented reasoning assessments.	L3 – Apply

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	**Prerequisites**	Basic understanding of patterns, sequences, numbers, alphabets, and simple problem-solving techniques and possess elementary analytical thinking skills and the ability to interpret information logically for drawing conclusions.	-	-
I	**Foundations of Logical Reasoning**	Introduction to Logical Reasoning, Analytical Thinking Techniques, Coding-Decoding, Letter and Number Series, Alphabet and Number Patterns. **Self-Learning Topics:** Direction Sense Test, Ranking and Ordering **Online Resources:** https://www.indiabix.com/logical-reasoning/ https://www.geeksforgeeks.org/reasoning/	**5**	**CO1**
II	**Blood Relations and Arrangement Problems**	Blood Relations, Family Tree Analysis, Linear Arrangement, Circular Arrangement, Seating Arrangement. **Self-Learning Topics:** Floor Arrangement, Relative Position Problems **Online Resources:** https://www.indiabix.com/verbal-reasoning/ https://www.hitbullseye.com	**5**	**CO2**
III	**Logical Puzzles and Analytical Reasoning**	Puzzle Solving Techniques, Scheduling Problems, Matching Puzzles, Grouping and Distribution Problems. **Self-Learning Topics:** Constraint-Based Reasoning, Analytical Caselets **Online Resources:**	**5**	**CO3**

		https://prepinsta.com/logical-reasoning/ https://www.careerride.com		
IV	**Statement-Based Reasoning**	Statements and Conclusions, Statements and Assumptions, Statements and Arguments, Course of Action, Cause and Effect, Assertion and Reason. **Self-Learning Topics:** Logical Reasoning Questions and Answers (IndiaBIX), Freshersworld – Jobs, Placement Papers, Aptitude and Interview Preparation **Online Resources:** https://www.indiabix.com/logical-reasoning/ https://www.freshersworld.com	**5**	**CO4**
V	**Non-Verbal and Visual Reasoning**	Mirror Images, Water Images, Paper Folding, Paper Cutting, Figure Completion. **Self-Learning Topics:** Embedded Figures, Pattern Recognition **Online Resources:** https://www.testbook.com/reasoning https://www.oliveboard.in	**5**	**CO5**
VI	**Advanced Placement Reasoning**	Data Sufficiency, Logical Deduction, Syllogisms, Venn Diagrams, Input-Output Logic. **Self-Learning Topics:** Critical Reasoning, Mixed Placement-Level Reasoning Sets **Online Resources:** https://www.placementpreparation.io https://www.geeksforgeeks.org/aptitude-for-placement	**5**	**CO6**
		Total Hours	**30**	
Note: Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.				

References:

1.R. S. Aggarwal, A Modern Approach to Verbal and Non-Verbal Reasoning, S. Chand Publications.
2.M. K. Pandey, Analytical Reasoning, BSC Publications.
3.Arun Sharma, How to Prepare for Logical Reasoning for CAT, McGraw Hill Education.

Online References:

Sr. No.	Website Name	URL
1	Logical Reasoning Questions and Answers (IndiaBIX)	Logical Reasoning Questions and Answers
2	GeeksforGeeks – Reasoning	GeeksforGeeks | 404
3	IndiaBIX – Verbal Reasoning	Hitbullseye: Transforming Education with Cutting Edge Technology

Assessment:

Internal Assessment Test (IAT) for 25 marks each:

- IA will consist of two compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, the remaining syllabus content must be covered in the IAT-II. 20 marks IAT-1, 20 marks IAT-2 Course Engagement Component: 5 marks (integrated within IAT-1 and IAT-2)

  Course Engagement (ARCS-based): The engagement component aims to motivate active student participation in classroom learning. Marks are awarded based on engagement aligned with the ARCS model. At the end of selected lectures, students will provide short reflective inputs—such as concepts they feel confident about or aspects that brought clarity or satisfaction. Each valid response earns 1 mark, up to a maximum of 5 marks.

- **Self Learning (SL):** Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%. This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

- Question Paper will comprise a total of seven questions. Question 1 will be compulsory (10 marks) and should cover the maximum contents of the syllabus (all 6 modules).
- The remaining Questions 2 to 7 will be mixed in nature, each carrying 10 marks. Each of these questions may have two parts: part (a) and part (b), each carrying 5 marks. Part (a) and part (b) of the same question must be from different modules. For example, if Q2 has part (a) from Module 3, then part (b) must be from any other module randomly selected from all the modules.
- A total of five questions need to be answered: Q1 (compulsory) + any four from Q2 to Q7. Total marks = 10 + (4 × 10) = 50 marks.